

Nonlinear Dynamics ! Koopman Operators

Consider a nonlinear autonomous dynamical sys.,

$$(*) \quad \dot{x} = F(x), \quad \text{s.t.} \quad x(0) = x_0 \in \mathbb{R}^d.$$

Here, $F: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a smooth, but typically nonlinear, map called the **dynamical law**.

If F is locally Lipschitz, meaning that there is a constant $L > 0$ s.t. $|F(x) - F(y)| \leq L|x - y|$ for all x, y sufficiently close to x_0 , then there is some $T > 0$ s.t. the solution to $(*)$ exists and is unique for all $0 \leq t \leq T$. It satisfies

$$(**) \quad x(t) = x_0 + \int_0^t F(x(s)) ds, \quad 0 \leq t \leq T.$$

We also consider discrete-time systems,

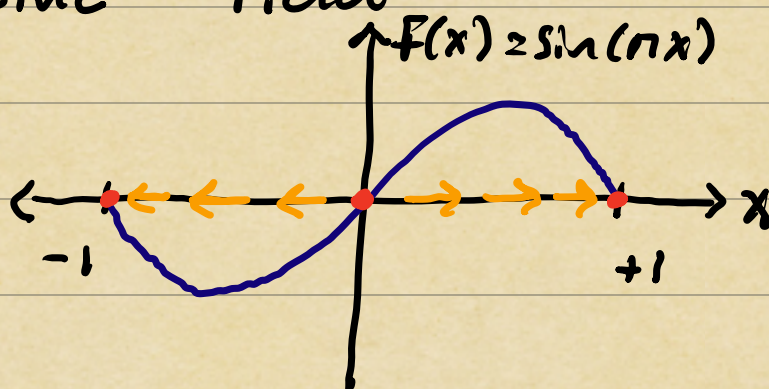
$$x_{k+1} = G(x_k), \quad k = 0, 1, 2, \dots$$

Geometric Approach

For low-dimensional dynamical systems, we can study solutions to (*) by plotting the vector field $F: \mathbb{R}^d \rightarrow \mathbb{R}^d$ around x_0 .

Example. 1D "Sine" Field

$$\dot{x} = \sin(\pi x)$$
$$x(0) = x_0 \in [-1, 1]$$



The dynamical system has 3 fixed points, where $F(x) = \sin(\pi x) = 0$. If $x_0 = -1, 0$, or 1 , then $x(t) = x_0$ for all $t \geq 0$! In between fixed points, the sign of $F(x) = \sin(\pi x)$ determines the direction of flow.

Q: Which fixed points "attract" nearby initial conditions? Which fixed points "repel"?

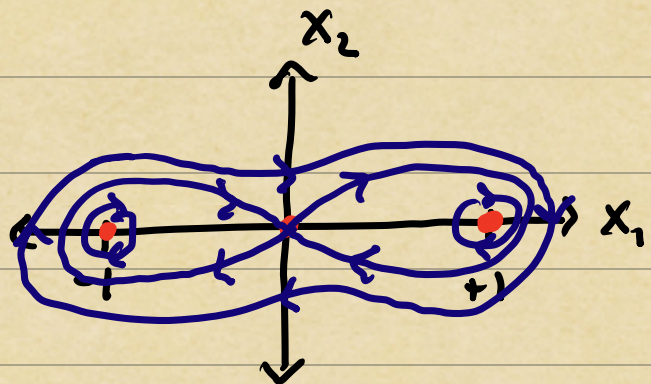
Q: What property of F determines this stability?

$\Rightarrow$ The slope of F at each fixed point determines if the point attracts or repels nearby initial conditions.

The geometric approach to nonlinear dynamics organizes the state space around the fixed points of F and their stability properties. The key idea is that the dynamics near a generic fixed point are approximately linear!

Example. Duffing Oscillator

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_1 - x_1^3\end{aligned}$$



Fixed Points: $x_1 = x_2 = 0$, $x_1 = \pm 1$ and $x_2 = 0$

Q: How can we determine the dynamics near the fixed points of F ?

By multivariate Taylor's theorem,

$$F(\underline{x}) = \cancel{F(\underline{x}_*)} + \overset{=0}{\text{Jacobian}} DF(\underline{x}_*)(\underline{x} - \underline{x}_*) + o(|\underline{x} - \underline{x}_*|) \\ \approx DF(\underline{x}_*)$$

For $F(x_1, x_2) = \begin{bmatrix} x_2 \\ x_1 - x_1^3 \end{bmatrix}$, the Jacobian is

$$DF(x_1, x_2) = \begin{bmatrix} 0 & 1 \\ 1 - 3x_1^2 & 0 \end{bmatrix}.$$

Evaluating at the fixed points, we get

$$\underline{x}_1 = \pm 1, \underline{x}_2 = 0$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} \approx \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\underline{x}_1 = \underline{x}_2 = 0$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} \approx \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\lambda_{\pm} = \pm \sqrt{2} i$$

$$\lambda_{\pm} = \pm 1$$

$\Rightarrow$ The dynamics at the origin have one stable and one unstable direction.

$\Rightarrow$ The dynamics at the points $x_1 = \pm 1, x_2 = 0$ are "centers," meaning there are periodic trajectories orbiting these fixed points.

$\Rightarrow$ Since there are no other fixed points and trajectories cannot intersect (by uniqueness, since F is locally Lipschitz on every compact subset of $\mathbb{R}^2$), we can reconstruct the qualitative image of the phase-space trajectories by smoothly "gluing" together the local dynamics.